Build guide – how the dashboard was made, step by step

This guide lets anyone rebuild the dashboard from the Kaggle files. See INSTALL.md to install Power BI Desktop and download the data first.

1. Load the data

Home → Get data → Text/CSV, then **Transform Data**. Load these 7 files and rename the queries:

CSV file	Query name
olist_orders_dataset.csv	Orders
olist_order_items_dataset.csv	OrderItems
olist_customers_dataset.csv	Customers
olist_sellers_dataset.csv	Sellers
olist_products_dataset.csv	Products
olist_order_reviews_dataset.csv	Reviews
product_category_name_translation.csv	CategoryTranslation

Payments and geolocation are not needed for this analysis.

Tip: if dates or decimals look wrong, set the column locale to **English (United States)** (right-click the column → Change Type → Using Locale).

2. Clean the data in Power Query

Orders

1. Filter `order_status` = `delivered`.
2. Filter `order_delivered_customer_date` to remove `(null)`.
3. **Add Column → Custom Column**, three times:

```
DelayDays = Duration.Days(Date.From([order_delivered_customer_date]) -
Date.From([order_estimated_delivery_date]))
DeliveryStatus = if [DelayDays] > 0 then "Late" else "On time"
DeliveryDays = Duration.Days([order_delivered_customer_date] - [order_purchase_timestamp])
```

Set `DelayDays` and `DeliveryDays` to Whole Number and `DeliveryStatus` to Text.

Reviews

1. Keep only `order_id` and `review_score`.
2. **Home → Group By** `order_id`, operation **Average** of `review_score`, new column `ReviewScore` (some orders have more than one review).

CategoryTranslation

1. **Home → Use First Row as Headers**.

Then **Home → Close & Apply**.

3. Build the model (Model view)

Delete any relationship Power BI created automatically, then create these. All are **many-to-one (*:1)** with **single** filter direction:

From (many)	To (one)
OrderItems[order_id]	Orders[order_id]
OrderItems[product_id]	Products[product_id]
OrderItems[seller_id]	Sellers[seller_id]
Orders[customer_id]	Customers[customer_id]
Reviews[order_id]	Orders[order_id]

Products[product_category_name]	CategoryTranslation[product_category_name]

The result is a star schema with **Orders** in the center.

4. DAX columns

Modeling → New column (in the table shown in brackets):

```
// Orders
Purchase Month = DATE(YEAR(Orders[order_purchase_timestamp]), MONTH(Orders[order_purchase_timestamp]),
1)
CarrierDays = IF(ISBLANK(Orders[order_delivered_carrier_date]), BLANK(),
INT(Orders[order_delivered_customer_date] - Orders[order_delivered_carrier_date]))
SellerShipping = IF(Orders[order_delivered_carrier_date] > MAXX(RELATEDTABLE(OrderItems),
OrderItems[shipping_limit_date]), "Seller shipped late", "Seller shipped on time")

// Sellers
Seller = "Seller " & LEFT(Sellers[seller_id], 6)
```

5. DAX measures

Modeling → New measure (all stored in Orders):

```
Total Orders = COUNTROWS(Orders)
Late Orders = CALCULATE(COUNTROWS(Orders), Orders[DeliveryStatus] = "Late")
Late % = DIVIDE([Late Orders], [Total Orders])
Avg Review = CALCULATE(AVERAGE(Reviews[ReviewScore]), Orders)
Avg Delivery Days = AVERAGE(Orders[DeliveryDays])
Avg Carrier Days = AVERAGE(Orders[CarrierDays])
Late Share from Late Shipping = DIVIDE(CALCULATE([Late Orders], Orders[SellerShipping] = "Seller
shipped late"), [Late Orders])

// Card labels with thousands separator
Total Orders Display = FORMAT([Total Orders], "#,0", "en-US")
Late Orders Display = FORMAT([Late Orders], "#,0", "en-US")

// Versions used by seller and category charts:
// the filter must flow from OrderItems back to Orders, only inside these measures
Total Orders (Items) = CALCULATE([Total Orders], CROSSFILTER(OrderItems[order_id], Orders[order_id],
Both))
Late Orders (Items) = CALCULATE([Late Orders], CROSSFILTER(OrderItems[order_id], Orders[order_id],
Both))
Late % (Items) = DIVIDE([Late Orders (Items)], [Total Orders (Items)])
```

Format `Late %`, `Late % (Items)` and `Late Share from Late Shipping` as percentages.

6. Report filter

In the **Filters** pane, under **Filters on all pages**, add `Orders[Purchase Month]` → *is on or after* 1/1/2017 (2016 has very few orders and creates false peaks).

7. Pages

Canvas size 1920 × 1080. Each page has the same title text box and 5 cards (Total Orders Display, Late Orders Display, Late %, Avg Review, Avg Delivery Days), plus a page navigator (**Insert → Buttons → Navigator → Page navigator**).

Page 1 – Overview

Visual	Fields
Line chart "Share of late orders by month"	X: Purchase Month · Y: Late %
Column chart "Late deliveries cut the average review score almost in half"	X: DeliveryStatus · Y: Avg Review

Page 2 – Where

Visual	Fields	Visual filter
Bar chart by customer state	Customers[customer_state] · Late %	–
Bar chart by seller state	Sellers[seller_state] · Late % (Items)	Total Orders (Items) ≥ 500
Bar chart by	CategoryTranslation[product_category_name_english] ·	Total Orders (Items) ≥ 500

category	Late % (Items)	
Bar chart top sellers	Sellers[Seller] · Late Orders (Items)	Seller: Top N = 10 by Late Orders (Items)

Page 3 – Why and what to do

Visual	Fields
Column chart "Late orders spend 27 days with the carrier vs 7.5 on time"	X: DeliveryStatus · Y: Avg Carrier Days
Column chart "When the seller ships late, 21% of orders arrive late (vs 5%)"	X: SellerShipping · Y: Late %
Card	Late Share from Late Shipping
Text box	The four recommendations

8. Formatting choices

- Chart titles state the conclusion, not just the measure.
- One accent color (orange) for "late", grey for "on time".
- Data labels on, X-axis values off on bar charts, white card backgrounds with rounded borders.